Lesson 2: Twist It

Year 1/2 STEAM | Unit: Stronger Together: Paper Makers | CRT plan | 45 minutes

No preparation needed. Everything is collected with student helpers during the lesson. **No screens** are used. All recorded work is glued into **STEAM journals**, which are on the shelves as you walk in the door.

Learning Intention and Success Criteria

Learning Intention:
To find out how twisting strips of paper together makes a stronger rope.

Success Criteria:
I can:

- twist paper strips together to make a rope.
- test how many books my rope can lift.
- explain how my rope is different from one strip.

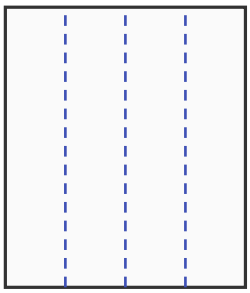
Read these aloud at the start and come back to them at the end.

What you need

Item	How many	Used for
A4 scrap or copy paper	2 sheets per pair, plus 1 per student	Making strips, and 1 sheet for recording
Small, thin books (readers or small picture books)	4 per pair, similar size	The weights for the lift test
Sticky tape	4 short pieces per pair	Taping the ends of the rope
Glue sticks	1 per pair	Gluing the record into journals
STEAM journals	1 per student	Shelves by the door

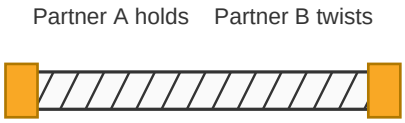
How the test works

1. Make strips



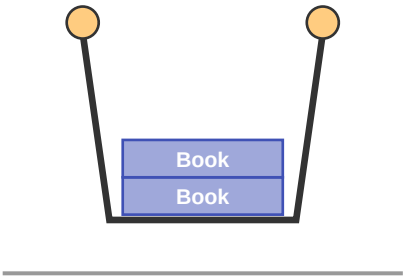
Fold, then tear on the folds
= 4 long strips

2. Twist 3 strips into a rope



Count to 20 while twisting
Tape both ends (gold)

3. Lift test



Lift just above the table
Add 1 book at a time

- **Make strips:** fold an A4 sheet in half the long way, then in half again. Open it and tear along the 3 folds. This makes 4 long strips.
- **Starter tear:** on every strip, make one tiny tear at the edge, in the middle, about as long as a fingernail. Every strip gets the same tiny tear so the test is fair.
- **Make a rope:** hold 3 strips together. Partner A holds one end tightly, Partner B twists the other end round and round while the pair counts to 20. Tape both ends so it does not unwind.
- **Lift test:** slide the strip (or rope) under a book. Hold one end in each hand and lift the book **just above the table**. If it holds, put it down, add another book on top and try again.
- The score is the **number of books lifted** before it tears. Test 1 single strip first, then the rope.

Key words

strip, twist, rope, strong, tear, together, test, predict

Lesson sequence

Time	Step	What to do and say
0 to 3 min	Settle	Students sit on the floor facing you. Ask: *"Last week, how did we make paper stronger?"* (By folding and rolling it.) Read the Learning Intention and Success Criteria aloud.
3 to 10 min	CIC hook	Tear one long strip from a sheet of paper. Ask: *"Could this one strip lift a book?"* Thumbs up or down. Make a tiny starter tear in the middle of the edge, slide it under one book and lift. It rips! Say: *"Paper is weak... or is it? Watch."* Tear 3 more strips, give each the same tiny tear, then ask a student to hold one end while you twist the other end as the class counts to 20. Tape both ends. Lift the same book. It holds! Ask: *"It is the same paper, with the same tears. What changed?"*
10 to 14 min	Explain the challenge	Tell them: *"Today you are rope makers. You will test 1 strip, then a rope made of 3 strips."* Show how to fold and tear 4 strips from one sheet. Show the tiny starter tear (fingernail size). Show how to lift: just above the table, add 1 book at a time. Each pair predicts how many books the rope will lift. Draw the recording layout (below) on the board.
14 to 30 min	Make and test in pairs	Helpers hand out 2 sheets, 4 pieces of tape and 4 books per pair. Pairs make 4 strips. Test 1 strip first and count the books. Then twist the other 3 strips into a rope and test it. Partners swap jobs: one lifts while the other adds books and counts. Roam with the questions below.
30 to 38 min	Record	Give each student 1 sheet. They fold it in half and draw the single strip and the rope on the front, write how many books each lifted, and circle the stronger one. Helpers hand out journals. Students glue the folded sheet into their journal and write the date and the title **Twist It**.
38 to 45 min	Share and pack up	3 or 4 pairs share using the sentence stem on the board. Finish with the key idea (below). Pack up: torn strips into recycling, books back, journals onto the shelves by the door. Ropes can go home or into recycling.

Draw this on the board

	One strip	Rope (3 strips)
Picture	(draw it)	(draw it)
Books lifted	___ books	___ books

Circle the stronger one.

Sentence stem: **The rope was stronger because _____.**

Questions to ask while roaming

- What happened to the tiny tear when you lifted the single strip?
- Why didn't the tear rip the rope?
- Is the rope made of anything new, or just paper?
- What do you think would happen with 4 strips, or with more twists?

Support and extension

Who	What to do
Support	Hold the end of the rope while the pair twists, and count the twists with them. Let them record with drawings and numbers only. Pair a confident counter with a student who needs help counting.
Extension	Fast finishers (often Year 2) make a second rope to compare: 2 strips against 4 strips, or a loose rope (5 twists) against a tight rope (30 twists). Which lifts more books? Write one sentence explaining why.

Key idea to finish on: One strip on its own tears easily. Twisted together, the strips help each other, so the **rope is stronger** than any strip by itself. Same paper, combined in a new way. Next week we will find out what happens when we **weave** paper strips together.

Notes for the teacher

- Most single strips rip on the first or second book. Most ropes lift 3 books or more. Any result is fine: the comparison is the learning.
- Keep the starter tear tiny. If a student makes a big tear, have them tear a fresh strip.
- If a rope unwinds, the ends were not taped tightly. Re-twist and re-tape.
- Lifting happens over the table, just above the surface, so books do not fall far.

- Please leave a short note about how far the class got and any pairs who were absent.

Curriculum links

- **Science VC2S2U05:** materials can be combined in a variety of ways for particular purposes; objects can have different properties from the materials they are made from.
- **Design and Technologies VC2TDE2C04:** explore the characteristics and properties of materials used to create designed solutions.
- **Maths:** counting to 20 (twists), counting a collection, comparing quantities.